UQFF

Author: Daniel T. Murphy

Session: *76 (March 2026)*

UQFF

`n = [int]# UQFF "PAPER_{0:D3}" -f [int]# UQFF` n = [int]# UQFF PAPER_276

Andromeda Friedmann-UQFF Gravity Coupling: H(z)■t Expansion Term and H_UQFF Near-Unity Resonance

Author: Daniel T. Murphy **Session:** 76 (March 2026) **Module:** ANDROMEDA*UQFF*MODULE.cpp (M31 Master, UQFF 2.0) **WOLFRAM_TERM:** `AndromedaUQFF:g_exp=G*M/r^2*H(z)*t; H(z)=H0*Sqrt[Om_m*(1+z)^3+Om_L]/Mpc2m; H_UQFF=H(z)*t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as g*expansion = (G■M/r■) ■ H(z) ■ t. For Andromeda's blueshift redshift z = -0.001, the Friedmann H(z) equals 69.969 km/s/Mpc = 2.269■10?■8 s?■. Evaluated at the Hubble timescale t = t*H = 4.352■10■7 s, the dimensionless Friedmann-UQFF Resonance Coefficient H*UQFF = H(z)■t*H ■ **0.987** ■ a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the M*visible/M*DM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

1. Background

The UQFF framework computes total gravitational acceleration g*total(r,t) as a sum of Newtonian, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER*273■275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter H(z). This is significant because over a Hubble timescale, H(z)■t ■ 1 ■ implying that expansion-mediated gravitational coupling is of the same order as the base Newtonian term.

2. Friedmann H(z) Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard ?CDM cosmology:

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H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}
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For Andromeda parameters: H0 = 70.0 km/s/Mpc, O*m = 0.3, O*? = 0.7, z = -0.001.

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H(z=-0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}
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= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}
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```
= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}
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In SI units: H_SI = 69.969 ■ 10■ / 3.086■10■■ = **2.269■10?■8 s?■**

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

```
g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t
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This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t.

Numerical values for Andromeda:

- g_base = G■M/r■ = 6.674■10?■■ ■ 1.989■104■ / (1.04■10■■)■ = **1.227■10?■■ m/s■**
- At t = 1 Gyr (3.156■10■6 s): g_expansion = 1.227■10?■■ ■ 2.269■10?■8 ■ 3.156■10■6 = **8.79■10?■■ m/s■**
- At t = t*Hubble (4.352■10■7 s): g*expansion = 1.227■10?■■ ■ 0.987 = **1.211■10?■■ m/s■**

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3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

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H_{\text{UQFF}} = H(z) \times t_H
```

where t_H = 4.352■10■7 s (Hubble time ■ 13.8 Gyr).

For Andromeda (z = -0.001):

```
H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}
```

3.2 Near-Unity Discovery

H_UQFF ■ 0.987 ■ **1** is a remarkable result. It means:

> *Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the base Newtonian term** ■ nearly doubling g_base.*

This near-unity value is not coincidental. In a flat ?CDM universe (O*m + O*? = 1):

```
H_{\text{UQFF}} = H_0 \times t_H \approx 1.0
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because H0 ■ t_H is the dimensionless Hubble number ■ 0.96■1.0 for observationally consistent H0.

New UQFF Constant: H*UQFF = H(z)*■*t*H ■ **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H0	70.0 km/s/Mpc
O_m	0.3

Parameter	Value
O_?	0.7
z (Andromeda)	-0.001
H(z)	2.269■10?■8 s?■
t_H	4.352■10■7 s
H_UQFF	**0.987** (~1)
g*expansion(t*H)	1.211■10?■■ m/s■
g_base	1.227■10?■■ m/s■
Ratio g*exp/g*base	98.7%

3.3 Blueshift Sensitivity

For blueshift z < 0: H(z) is slightly **lower** than H0 (matter term O*m■(1+z)■ is suppressed below O*m for z < 0), meaning:

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H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)
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For Andromeda (z = -0.001): H*UQFF = 0.987 vs flat-universe H*UQFF0 = 0.9985. The blueshift suppresses H_UQFF by 0.15%.

This is the inverse of the redshift behaviour (z > 0 ? H(z) > H0 for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

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4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

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a_{\text{dust}} = \frac{\rho_{\text{dust}} \cdot v_{\text{orbit}}^2}{c^2 \cdot
\rho_{\text{mean}}} \cdot g_{\text{base}}
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where ?*mean = M/V*fluid = 1.989■104■ / 106■ = 1.989■10?■8 kg/m■.

Numerical value for Andromeda:

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a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times
1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}
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= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times
1.227 \times 10^{-10}
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```
= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19}
\text{ m/s}^2
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At ~9 orders below g_base, the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

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5. M*visible / M*DM_mass Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

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M_{\text{visible}} = (1 - f_{\text{DM}}) \times M
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M_{\text{DM,mass}} = f_{\text{DM}} \times M
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For Andromeda defaults: M = 1.989■104■ kg, f_DM = 0.80:

M_visible = 0.20 ■ 1.989■104■ = **3.978■104■ kg** (20% visible baryons)

M*DM*mass = 0.80 ■ 1.989■104■ = **1.591■104■ kg** (80% dark matter)

These are tracked in `updateCache()` and exported via `exportState()` (params 26■32), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

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6. Updated ANDROMEDA*UQFF*MODULE.cpp g_total

The full UQFF 2.0 equation for Andromeda with all PAPER_273■276 terms:

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g_{\text{total}}(r, t) = \left[ g_{\text{grav}} + U_{g,\text{sum}}^{(26)} +
\Lambda\text{-term} + g_Q + g_L + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) +
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Term magnitudes at t=0 (current epoch):

Term	Value (m/s■)	Paper
g_grav = G■M/r■	1.227■10?■■	baseline
Ug_sum (26 layers)	6.380■10??	26-layer Triadic
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g_quantum	~2■10?■6	HUP
g_Lorentz	~2■10?■6	IGM EM
g_fluid	~8■10?■■	IGM buoyancy
F*res(?*HI) at t=0	1■10?■■	PAPER_274
g_DM	1.356■10?■■	PAPER_275
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Note: g*expansion grows linearly with t, dominating over g*grav at t = t_H.

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7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`:

`H0` = 70.0 km/s/Mpc

`Omega_m` = 0.3

`Omega_Lam` = 0.7

`Mpc_to_m` = 3.086■10■■

`rho_dust` = 1■10?■■ kg/m■

`M_visible` = (1■f_DM)■M

`M_DM_mass` = f_DM■M

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8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

- **Friedmann-UQFF Expansion Coupling**: g*expansion = g*base ■ H(z) ■ t. At t = t*H, adds 98.7% of g*base ■ near-gravitational doubling over cosmological time.
- **H_UQFF = 0.987**: New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship H0 ■ t_H ■ 1 in flat ?CDM. Andromeda's blueshift (z = -0.001) makes H_UQFF 0.15% below flat-universe value ■ a blueshift Friedmann suppression.
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where ?*mean* = *M/V*fluid = 1.989■104■ / 106■ = 1.989■10?■8 kg/m■.

Numerical value for Andromeda:

```
a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times
1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}
```

```
= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times
1.227 \times 10^{-10}
```

```
= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19}
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At ~9 orders below g_base, the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

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When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

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6. Updated ANDROMEDA*UQFF*MODULE.cpp g_total

The full UQFF 2.0 equation for Andromeda with all PAPER_273■276 terms:

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g_{\text{total}}(r, t) = \left[ g_{\text{grav}} + U_{g,\text{sum}}^{(26)} +
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g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times
\kappa_{\text{approach}}
```

Term magnitudes at t=0 (current epoch):

Term	Value (m/s■)	Paper
g_grav = G■M/r■	1.227■10?■■	baseline
Ug_sum (26 layers)	6.380■10??	26-layer Triadic
?-term	2.998■10?■6	cosmological
g_quantum	~2■10?■6	HUP
g_Lorentz	~2■10?■6	IGM EM
g_fluid	~8■10?■■	IGM buoyancy
F*res(?*HI) at t=0	1■10?■■	PAPER_274
g_DM	1.356■10?■■	PAPER_275
g_expansion at t=0	0	PAPER_276
a_dust	4.29■10?■?	PAPER_276
?_approach	1.001001	PAPER_273

Note: g*expansion grows linearly with t, dominating over g*grav at t = t_H.

---

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`:

`H0` = 70.0 km/s/Mpc

`Omega_m` = 0.3

`Omega_Lam` = 0.7

`Mpc_to_m` = 3.086■10■■

`rho_dust` = 1■10?■■ kg/m■

`M_visible` = (1■f_DM)■M

`M_DM_mass` = f_DM■M

---

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

- **Friedmann-UQFF Expansion Coupling**: g*expansion = g*base ■ H(z) ■ t. At t = t*H, adds 98.7% of g*base ■ near-gravitational doubling over cosmological time.
- **H_UQFF = 0.987**: New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship H0 ■ t_H ■ 1 in flat ?CDM. Andromeda's blueshift (z = -0.001) makes H_UQFF 0.15% below flat-universe value ■ a blueshift Friedmann suppression.
- **M_visible/M_DM_mass cascade + dust drag**: Explicit DM split bookkeeping (updateCache) and ISM dust ram-pressure (a_dust ■ 4■10?■? m/s■).

Andromeda Friedmann-UQFF Gravity Coupling: H(z)■t Expansion Term and H_UQFF Near-Unity Resonance

Author: Daniel T. Murphy **Session:** 76 (March 2026) **Module:** ANDROMEDA*UQFF*MODULE.cpp (M31 Master, UQFF 2.0) **WOLFRAM_TERM:** `AndromedaUQFF:g_exp=G*M/r^2*H(z)*t; H(z)=H0*Sqrt[Om_m*(1+z)^3+Om_L]/Mpc2m; H_UQFF=H(z)*t_H~0.987 [PAPER_276]`

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as g*expansion = (G■M/r■) ■ H(z) ■ t. For Andromeda's blueshift redshift z = -0.001, the Friedmann H(z) equals 69.969 km/s/Mpc = 2.269■10?■8 s?■. Evaluated at the Hubble timescale t = t*H = 4.352■10■7 s, the dimensionless Friedmann-UQFF Resonance Coefficient H*UQFF = H(z)■t*H ■ **0.987** ■ a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the M*visible/M*DM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

---

1. Background

The UQFF framework computes total gravitational acceleration g*total(r,t) as a sum of Newtonian, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER*273■275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter H(z). This is significant because over a Hubble timescale, H(z)■t ■ 1 ■ implying that expansion-mediated gravitational coupling is of the same order as the base Newtonian term.

---

2. Friedmann H(z) Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard ?CDM cosmology:

```
H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}
```

For Andromeda parameters: H0 = 70.0 km/s/Mpc, O*m = 0.3, O*? = 0.7, z = -0.001.

```
H(z=-0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}
```

```
= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}
```

```
= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}
```

In SI units: H_SI = 69.969 ■ 10■ / 3.086■10■■ = **2.269■10?■8 s?■**

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

```
g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t
```

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t.

Numerical values for Andromeda:

- g_base = G■M/r■ = 6.674■10?■■ ■ 1.989■104■ / (1.04■10■■)■ = **1.227■10?■■ m/s■**
- At t = 1 Gyr (3.156■10■6 s): g_expansion = 1.227■10?■■ ■ 2.269■10?■8 ■ 3.156■10■6 = **8.79■10?■■ m/s■**
- At t = t*Hubble (4.352■10■7 s): g*expansion = 1.227■10?■■ ■ 0.987 = **1.211■10?■■ m/s■**

---

3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

```
H_{\text{UQFF}} = H(z) \times t_H
```

where t_H = 4.352■10■7 s (Hubble time ■ 13.8 Gyr).

For Andromeda (z = -0.001):

```
H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}
```

3.2 Near-Unity Discovery

H_UQFF ■ 0.987 ■ **1** is a remarkable result. It means:

> *Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to* ***98.7% of the base Newtonian term*** *■ nearly doubling g_base.*

This near-unity value is not coincidental. In a flat ?CDM universe (O*m + O*? = 1):

```
H_{\text{UQFF}} = H_0 \times t_H \approx 1.0
```

because H0 ■ t_H is the dimensionless Hubble number ■ 0.96■1.0 for observationally consistent H0.

New UQFF Constant: H*UQFF = H(z)*■*t*H ■ **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H0	70.0 km/s/Mpc
O_m	0.3
O_?	0.7
z (Andromeda)	-0.001
H(z)	2.269■10?■8 s?■
t_H	4.352■10■7 s
H_UQFF	**0.987** (~1)
g*expansion(t*H)	1.211■10?■■ m/s■
g_base	1.227■10?■■ m/s■
Ratio g*exp/g*base	98.7%

3.3 Blueshift Sensitivity

For blueshift z < 0: H(z) is slightly **lower** than H0 (matter term O*m*■*(1+z)*■ *is suppressed below O*m for z < 0), meaning:

```
H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)
```

For Andromeda (z = -0.001): H*UQFF = 0.987 vs flat-universe H*UQFF0 = 0.9985. The blueshift suppresses H_UQFF by 0.15%.

This is the inverse of the redshift behaviour (z > 0 ? H(z) > H0 for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

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UQFF computed: MUGE buoyancy ratio U*bi/F*U = [SSq]■?■r■/GM = 5.7e-1■5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s■ at r_ISCO.
